MARK SCHEME · TEACHER COPY

12BR · Topic 20: Further Programming

Cambridge A Level Computer Science 9618

Paper 3 Advanced Theory

Total marks: 37

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 37 marks

Question 1 · Oct/Nov 2025 Paper 31, original Q11 (a),(b) [8 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>One mark per mark point
MP1 Two correct attributes (<code>PatientID : INTEGER</code> and <code>Doctor : STRING</code>)
MP2 <code>SetTreatments(…)</code> and <code>SetMedications(…)</code> seen
MP3 … and appropriate parameters, with string data types
MP4 <code>GetPatientID()</code> and <code>GetDoctor()</code> seen
MP5 <code>SetDateSeen(…)</code> and <code>GetDateSeen</code> fully correct with appropriate parameter and correct data type in setter.

Appointment

<code>DateSeen : DATE</code>
PatientID : INTEGER
Doctor : STRING
<code>Treatments : STRING</code>
<code>Medications : STRING</code>

SetDateSeen(NewDate : DATE)
<code>SetPatientID(PatientNumber : INTEGER)</code>
<code>SetDoctor(DoctorID : STRING)</code>
SetTreatments(NewTreatments : STRING)
SetMedications(NewMedications : STRING)
GetDateSeen()
GetPatientID()
GetDoctor()
<code>GetTreatments()</code>
<code>GetMedications()</code></td><td>5</td></tr>
<tr><td>1(b)(i)</td><td>Encapsulation</td><td>1</td></tr>
<tr><td>1(b)(ii)</td><td>One mark for each mark point (Max 2)
MP1 Inheritance is where a derived class takes properties / behaviours attributes / methods
MP2 … of a parent / super class / base class
MP3 The attributes / methods / properties taken from the parent / super class / base class can also be extended / copied / used / changed / overwritten / overridden in the subclass.</td><td>2</td></tr>
</table>

Total: 37 marks

Question 2 · May/June 2026 Paper 32, original Q11 (a),(b) [9 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>2(a)</td><td>One mark per mark point (Max 5)
MP1 Correct <code>CLASS Node</code> and <code>ENDCLASS</code> statements
MP2 Correct attribute declaration section before the constructor
MP3 Correct statement to begin and end the constructor as <code>PUBLIC PROCEDURE NEW ... ENDPROCEDURE</code>
MP4 Two parameters identified in the constructor with appropriate names and data types.
MP5 Values passed using the constructer are assigned correctly – allow use of setters.

Example algorithm

<pre>CLASS Node
 PRIVATE Content : STRING
 PRIVATE Pointer : INTEGER

 PUBLIC PROCEDURE NEW(NewContent : STRING,
 NewPointer : INTEGER)
 Content ← NewContent
 Pointer ← NewPointer
 ENDPROCEDURE
ENDCLASS</pre></td><td>5</td><td>MP1 <code>Node</code> is given in the question, so must be used.
MP2 must use <code>Content</code> and <code>Pointer</code> as <code>STRING</code> and <code>INTEGER</code>, respectively.
MP4 any names can be used except <code>Content</code> and <code>Pointer</code>.
MP5 the candidate's parameters from MP4 must be assigned to the attributes from MP2.</td></tr>
<tr><td>2(b)</td><td>One mark per correct answer (Max 4)

<table><tr><th>OOP Term</th><th>Description</th></tr><tr><td>setter</td><td>A method used to change values in private attributes in a class.</td></tr><tr><td>instance(s) // object(s)</td><td>An occurrence of a class during the execution of a program.</td></tr></table></td><td>4</td><td></td></tr>
</table>

aggregation	One class/object contains another
	// Hierarchy of nested classes/objects
	// Process by which one class can contain other classes
	// Represents a “has a” relationship between classes.

inheritance	Deriving new classes/objects on existing classes/objects
	// Process in which the methods and data from one class, a superclass or base class, are copied to another class, a derived class
	// Represents a hierarchy of derived classes
	// Represents an “is a” relationship between classes.

Question 3 · Oct/Nov 2025 Paper 32, original Q12 (a),(b) [8 marks]

Question	Answer	Marks
3(a)	**One** mark per mark point MP1 Two correct attributes (`DateOfBirth : Date`) and priority with a sensible name and integer data type MP2 `SetPatientID(…)` and `SetDoctorID(…)` seen MP3 … and appropriate parameters, with string data types MP4 `GetPatientID()` and `GetDateOfBirth()` seen MP5 Priority setter and getter fully correct with name matching Priority attribute, with appropriate parameter and correct data type. **Patient**	5

	PatientID : STRING Name : STRING **DateOfBirth : DATE** **Priority : INTEGER** DoctorID : STRING	
	SetPatientID(PatientNumber : STRING) SetName(FullName : STRING) SetDateOfBirth(DOB : DATE) **SetPriority(Urgency : INTEGER)** **SetDoctorID(DocID : STRING)** **GetPatientID()** GetName() **GetDateOfBirth()** **GetPriority()** GetDoctorID()	
3(b)(i)	Instance // instantiation	**1**
3(b)(ii)	**One** mark for each mark point MP1 Polymorphism is when methods with the same name are redefined / behave differently … MP2 … in derived / inherited classes / subclasses	**2**

Question 4 · May/June 2026 Paper 31, original Q11 (a),(b),(c) [8 marks]

Question	Answer	Marks	Guidance
4(a)	**One** mark for each correctly completed clause (**Max 3**) (25) customer(betty). (26) owns(large_tv, betty). (27) wants(bluray_player, betty).	3	Accept facts in any order. The device names have been given in clauses 02 and 04. Responses must match these.

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Total: 37 marks

4(b)	`(Owner =) petrov, susan, sammy`	**1**	Can be in any order

4(c)	**One** mark per mark point (**Max 4**)
	MP1 `device(D)`
	MP2 `customer(A)`
	MP3 `owns(D, A).`
	MP4 All logic conditions / punctuation correct (accept , for `AND`)
	
	Example answers
	
	`may_buy_device(D, A)`
	`  IF device(D) AND customer(A) AND NOT owns(D, A).`
	
	`may_buy_device(D, A)`
	`  IF device(D), customer(A), NOT owns(D, A).`
		4	Allow MP1, MP2 and MP3 in any order.
			
			Accept \+ for NOT operator as Swish Prolog.

Question 5 · Oct/Nov 2025 Paper 33, original Q5 [4 marks]

Question	Answer	Marks
5	**One** mark per correct answer (**Max 4**)	**4**

OOP Term	Purpose
Getter	A method that accesses the value of a property
Setter	A method that changes the value of a property
Object	**An instantiation / instance of a class**
Method	**A programmed function / procedure / subroutine / subprogram defined as part of a class**

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